

Towards Well-Calibrated AutoML: A Theoretical Analysis based on Ensemble Diversity

Salomey Osei¹, Andrés R. Masegosa², and Antonio D. Masegosa^{1,3}

¹ DeustoTech, Faculty of Engineering, University of Deusto, 48007 Bilbao, Spain

² Department of Computer Science, Aalborg University, 2450 Copenhagen SV, Denmark

³ Ikerbasque, Basque Foundation for Science, 48009 Bilbao, Spain

Abstract. Automated Machine Learning (AutoML) streamlines complex machine learning processes, making advanced analytical capabilities accessible to broader audiences. However, most existing AutoML frameworks emphasize predictive accuracy, often neglecting the critical aspect of probability calibration, essential in high-stakes decision-making domains. Miscalibrated probabilities can significantly compromise uncertainty estimates, leading to poor decision outcomes in areas such as medical diagnostics, autonomous driving, and financial risk management. This paper investigates the potential of ensemble-based AutoML methods to naturally improve probability calibration by implicitly increasing the so-called *diversity* of the models defining the ensemble. Specifically, we examine AUTO-SKLEARN, a leading ensemble-based AutoML framework, analyzing how its components—Base Learners, Meta-learning, Bayesian optimization, and Caruana ensemble construction—contribute to enhancing probability calibration by inducing better *diversity*. Our analysis reveals that ensemble-based AutoML inherently supports mechanisms beneficial for producing well-calibrated probability estimates. By uncovering the structural elements of ensemble AutoML that foster better uncertainty quantification, this work establishes empirical evidence for further research aimed at developing automated tools capable of consistently generating predictions with well-calibrated probability estimates in real-world applications.

Keywords: Automated Machine Learning (AutoML), AUTO-SKLEARN, Bayesian Optimization (BO), Diversity, Ensemble learning, Uncertainty quantification

1 Introduction

Automated Machine Learning (AutoML) has emerged as a transformative approach in modern data science, democratizing access to machine learning by automating traditionally challenging tasks such as model selection, hyperparameter tuning, and feature engineering [25]. By significantly reducing the required manual intervention and domain expertise, AutoML empowers a broader audience,

enabling even non-experts to leverage advanced machine learning capabilities effectively. Frameworks like AUTO-SKLEARN [4] and TPOT [18] exemplify this revolution, demonstrating competitive performance across diverse machine learning tasks through automated pipeline construction and hyperparameter optimization.

Despite these advancements, existing AutoML systems predominantly focus on maximizing predictive accuracy, often neglecting the quality of probabilistic outputs, crucial in many real-world scenarios [8]. In high-stakes decision-making domains, such as healthcare diagnostics, autonomous driving, or financial risk management, merely providing accurate predictions is insufficient. Decisions in these contexts rely heavily on accurate and reliable estimates of predictive uncertainty—probabilities that genuinely reflect the likelihood of predicted outcomes. For instance, in medical diagnosis, a well-calibrated probability means that among patients who receive a predicted probability of 70% for having a certain disease, exactly 70% indeed have that disease. Miscalibrated probabilities, therefore, can lead to either under- or over-estimation of risks, severely impacting decision quality and patient outcomes [17].

Despite the critical importance of calibrated probabilistic predictions, the literature on AutoML has scarcely addressed this aspect. Traditional AutoML frameworks are thereby missing opportunities for simultaneous optimization of predictive performance and probability calibration. This gap becomes particularly significant in high-stakes environments such as the ones mentioned above, where decisions not only require high accuracy but also demand reliable and trustworthy uncertainty estimates.

In this work, we aim to theoretically explore the potential of ensemble-based AutoML methods for providing well-calibrated probability estimates. We focus on ensemble AutoML methods because ensemble techniques have traditionally been recognized as powerful tools for improving probability calibration in classification tasks [12]. Unlike empirical evaluations that compare the effectiveness of different calibration techniques, our study does not seek to determine whether ensemble AutoML methods are more competitive than existing approaches. *Instead, we investigate whether these methods inherently possess mechanisms that are essential for producing well-calibrated probability estimates.*

More specifically, we analyze AUTO-SKLEARN, one of the most successful ensemble-based AutoML frameworks, to determine whether its automated ensemble-building process can naturally yield well-calibrated probability estimates. To achieve this, we build on recent theoretical analyses demonstrating that ensemble diversity plays a crucial role in probability calibration [16, 19]. We examine how the different stages of AUTO-SKLEARN such as Base Learners, Meta-learning, Bayesian optimization, and Caruana ensemble construction contribute to promoting diversity and, consequently, improving probability calibration. By understanding these mechanisms, we aim to provide theoretically grounded insights that can be used to develop novel AutoML methods that produces not only accurate predictions but also well-calibrated probability es-

timates, a key property for many applications involving high-stakes decision making.

The remainder of the paper is structured as follows. Sections 2 and 3 cover the background and related work, respectively. Then, Section 4 describes the theoretical framework used for analysis diversity in Ensemble based AutoML methods and its relation with well-calibrated probability estimates. The experimental framework used and the results obtained are depicted and analyzed in Section 5. Finally, the main conclusions of the paper are gathered in Section 6.

2 Background

AutoML is a fast growing field that seeks to automate the process of applying ML to real-world challenges. AutoML systems are intended to make ML more accessible to non-experts while improving the productivity and efficiency of experts. It has become an essential tool for democratizing ML by automating model selection, hyperparameter tuning, and pipeline optimization.

AutoML methods can be categorized based on the following ML processes: **i) Hyperparameter Optimization:** the tuning of hyperparameters in search of the best configuration for a model [22]; **ii) Neural Architecture Search:** the automatic design of the architecture of a neural network [14]; **iii) Combined Algorithm Selection and Hyperparameter Optimization:** the selection of the best hyperparameters and algorithm simultaneously [23]; **iv) Pipeline Optimization:** searching for the best combination of ML algorithms, hyperparameters, etc [18]; **v) Feature Engineering Automation:** automatically create and select the most important features for ML methods [10]; **vi) Ensemble Learning Automation:** automatic combination of multiple models to increase prediction performance [4, 13]; **vii) Transfer Learning in AutoML:** leverages pre-trained models and prior knowledge to adapt quickly to new tasks [27]; **viii) Evolutionary and Genetic Methods:** makes use of genetic programming and evolutionary methods for optimizing models and pipelines [11, 21]; **ix) Meta-Learning:** designing models that can learn new tasks efficiently using prior experience [24]; **x) End-to-End AutoML Systems:** unlike traditional ML, these are frameworks that are fully automatic and designed to handle the entire ML’s lifecycle [9].

Ensemble-based AutoML methods are the primary focus of our study because of their demonstrated ability to produce models with well-calibrated probability estimates. This calibration is of utmost importance in real-world applications where reliable uncertainty quantification is essential for informed decision-making. The strength of ensembles lies in their ability to combine multiple models, to reduce variance, and enhance robustness. This combination often results in a more accurate probability estimate compared to individual models. Our research aims to delve deeper into how these ensemble-based AutoML methods contribute to probability calibrations. We seek to understand the ways by which they achieve well-calibrated estimates. By this, we hope to enhance the uncer-

tainty quantification capabilities of these systems to lead to more reliable and interpretable AutoML models.

2.1 Ensemble AutoML and AUTO-SKLEARN

Ensemble AutoML is an approach within Automated Machine Learning (AutoML) centered on creating and combining multiple ML models, known as ensembles, to improve prediction performance. Ensemble methods, which leverages the strengths of various algorithms while reducing their individual weakness, with the aim of creating a more robust and accurate model than single-model methods. This technique, which is well-known in classical machine learning, is smoothly integrated into AutoML frameworks to improve automation and simplify model development. Ensemble AutoML systems train several models using different ML algorithms with unique hyperparameter settings. This diversity in models has been proven to improve uncertainty quantification, as combining multiple models reduces the variance of the uncertainty estimates and can lead to more reliable predictions [12]. The frameworks uses advanced optimization methods like Bayesian optimization to automate the process of hyperparameter tuning to reduce the need for manual intervention. After training, models are combined into an ensemble using methods like bagging [1], boosting [5], which has been found to enhance uncertainty estimation by leveraging the complementary strengths of each model [16, 19].

Among the various Ensemble AutoML techniques available, AUTO-SKLEARN has emerged as one of the most popular frameworks. Its popularity stems from a number of factors, including its ease of use, efficiency, and effectiveness for AutoML tasks. AUTO-SKLEARN is built on Scikit-learn, a widely used ML library. This integration assures familiarity for users who are already used to the scikit-learn’s interface, making AUTO-SKLEARN available to a wide range of users, from novices to seasoned practitioners. AUTO-SKLEARN has a robust ensemble learning mechanism that blends Bayesian optimization and meta-learning techniques. The approach provides high-performance outputs by combining the strengths of various models while minimizing their individual weaknesses. The framework simplifies model development by automating model selection and hyperparameter tuning. This automation allows users to focus on higher-level objectives rather than manually experimenting with algorithms and configurations. During the offline phase, AUTO-SKLEARN retrieves 38 meta-features from the data. These meta-features capture dataset characteristics including dimensionality, sparsity, and distribution qualities. The meta-learning stage employs these meta-features to "warm-start" Bayesian optimization by initializing it using machine learning framework configurations that are expected to perform well on new datasets. This warm-start approach speeds up optimization and improves model selection. Bayesian optimization is used to efficiently search the hyperparameter space and determine the best configurations for each model. By balancing exploration (searching for new configurations) and exploitation (refining promising configurations), Bayesian optimization ensures robust per-

formance across diverse datasets. In the following, we describe in more detail each of these phases:

- **Base-Learners:** Commonly known as weak learners, are essential components in ensemble learning, including AUTO-SKLEARN. These individual algorithms, while not necessarily strong predictors on their own, collectively contribute to a robust and high-performing ensemble model. The strength of this approach lies in leveraging diverse perspectives on the data to improve overall predictive accuracy. Each base-learner is trained with different initializations and on various subsets of the training data, ensuring a wide range of perspectives. The independence in their training processes allows each learner to contribute unique knowledge to the ensemble, helping to mitigate individual biases and reduce overall variance.
- **Meta-Learner:** Meta-learners in AUTO-SKLEARN enhances ensemble performance by combining outputs from multiple base-learners through averaging techniques or learning-based methods. This approach leverages the diversity and knowledge of base-learners to improve overall predictions. AUTO-SKLEARN implements meta-learning as a complementary step to Bayesian optimization, particularly to expedite hyperparameter search initialization for new datasets. During an offline phase, AUTO-SKLEARN trains on diverse datasets, extracting 38 meta-features that characterize specific datasets. These meta-features are then utilized by Bayesian optimization to suggest well-performing model configurations for new datasets, addressing the initial slowness of the optimization process. The meta-learner’s primary role is to estimate what are the most appropriate base-learner classifiers for ”warm-up” the Bayesian Optimization phase that it is described below.
- **Bayesian optimization:** rooted in Bayes theorem, offers an efficient solution for the complex ”black box” optimization problems, particularly when evaluating objective functions is computationally expensive [6]. This method mimics manual hyperparameter tuning but with enhanced efficiency, systematically refining parameters to identify optimal configurations. Bayesian optimization excels at proposing competitive hyperparameter candidates within a limited number of evaluations, thereby improving the performance of base-learners [2]. AUTO-SKLEARN employs Bayesian optimization not just for hyperparameter tuning but also for pipeline selection, efficiently navigating the complex landscape of configurations and enhancing overall system performance [2].
- **Caruana Ensemble:** Caruana’s ensemble method [3] aims to improve model performance by combining multiple base models. AUTO-SKLEARN adopts this strategy, leveraging ensembles’ superior performance over individual models. Rather than discarding equally performing models from Bayesian optimization, AUTO-SKLEARN employs automated ensemble construction. This approach mitigates overfitting risks and introduces diversity through varied models identified by Bayesian optimization. Implementing Caruana’s greedy ensemble approach, AUTO-SKLEARN constructs an ensemble of 50 models from the top-performing candidates.

Table 1: A summary of the 20 datasets used in the study, including the number of features (numerical and categorical), instances, missing values, and class distribution (majority and minority classes). These datasets were chosen based on many factors, including diversity and real-world representation

Dataset	Feat	Inst	Num. Feat	Cat. Feat	Missing Val	Classes	Maj Class	Min Class
Air Pressure SF	171	76,000	170	1	1,078,695	2	59,000	1,000
King Rook vs Pawn	37	3,196	0	37	0	2	1,669	1,527
Credit-G	21	1,000	7	14	0	2	700	300
Sylvine	21	5,124	20	1	0	2	2,562	2,562
Albert	79	425,240	26	53	2,734,000	2	212,620	212,620
Christine	1,637	5,418	1,599	38	0	2	2,709	2,709
Jasmine	145	2,984	8	137	0	2	1,492	1,492
KC1	22	2,109	21	1	0	2	1,783	326
Miniboone	51	130,064	50	1	0	2	93,565	36,499
Blood-Transf	5	748	4	1	0	2	570	178
Phoneme	6	5,404	5	1	0	2	3,818	1,586
Nomao	119	34,465	89	30	0	2	24,621	9,844
Numerai28.6	22	92,320	21	1	0	2	48,658	47,662
Bank-Mark	17	45,211	7	10	0	2	39,922	5,289
Adult	15	48,842	6	9	6,465	2	37,155	11,687
Amazon Emp	10	32,769	0	10	0	2	30,872	1,897
Car	7	1,728	0	7	0	4	1,210	65
CNAE-9	857	1,080	856	0	0	9	120	120
Fabert	801	8,237	800	1	0	6	1,927	502
Vehicle	19	846	18	1	0	5	218	199

3 Related Work

Automated Machine Learning (AutoML) has significantly streamlined the application of machine learning by automating key tasks, including algorithm selection, hyperparameter tuning, feature engineering, and pipeline optimization [4, 18]. Frameworks such as AUTO-SKLEARN [4] and TPOT [18] exemplify these advances, demonstrating strong predictive performance across diverse datasets and democratizing machine learning by reducing the barrier to entry for non-expert users [25].

Despite these significant advancements, the existing AutoML literature predominantly focuses on maximizing discriminative performance metrics, particularly accuracy and error rate. Typically, calibration—the alignment between predicted probabilities and actual outcome frequencies—is not explicitly incorporated into the optimization objectives of AutoML systems. This separation between model optimization and probability calibration poses significant limitations for applications demanding reliable probabilistic outputs.

Probabilistic calibration is essential in numerous real-world scenarios, especially those involving high-stakes decision-making. For instance, in medical diagnosis, financial risk assessment, or autonomous driving, accurately quantifying prediction uncertainty is critical for informed decision-making. Niculescu-Mizil and Caruana [17] emphasized that high accuracy alone does not guarantee good calibration, illustrating the necessity of explicitly optimizing for calibration qual-

ity. Recent studies further revealed that advanced models, including deep neural networks, frequently suffer from severe miscalibration, leading to overconfident predictions [8].

A notable exception is the AutoCP framework [26], which extends conformal prediction to automate the construction of prediction intervals that balance coverage guarantees and interval width. While AutoCP focuses on regression tasks, its principles—leveraging cross-validation and optimizing non-conformity scores—could theoretically extend to classification. However, its reliance on conformal prediction introduces computational overhead and risks over-conservatism, particularly in high-dimensional settings [20]. Other AutoML tools, such as TPOT or AUTO-SKLEARN, lack explicit mechanisms for uncertainty quantification (UQ), instead treating probabilistic outputs as byproducts of model architectures (e.g., logistic regression or Bayesian neural networks) without calibration safeguards [8].

Ensemble methods, widely recognized for their robustness and generalization capabilities, inherently provide avenues to improve calibration through diversity among constituent models [12, 16]. By leveraging the diverse predictive behaviors of multiple classifiers, ensemble methods can naturally enhance calibration performance. However, despite the well-established benefits of ensemble methods for calibration, their systematic exploration within AutoML frameworks has been minimal.

Specifically, in the context of ensemble AutoML, AUTO-SKLEARN stands out due to its successful track record, notably winning challenges such as the ChaLearn AutoML competitions. AUTO-SKLEARN constructs ensembles by automatically selecting and combining diverse pipelines optimized through Bayesian optimization and meta-learning [4]. Nevertheless, even this leading AutoML method primarily targets metrics such as accuracy or error rate, with probability calibration typically addressed separately, post-training. The theoretical underpinnings and potential of ensemble-based AutoML methods for inherently producing calibrated probability estimates remain insufficiently studied and poorly understood.

4 A Theoretical Framework for Diversity and Well-Calibrated Probability Estimates

Ensemble methods integrate multiple models to bolster predictive performance, and they also play a critical role in refining probability estimates. In this context, log-loss—a proper scoring rule—emerges as a key measure of how well-calibrated these probability estimates are: models that systematically over- or under-estimate probabilities incur higher log-loss.

A central driver of improved performance in ensemble learning is diversity among individual predictors. By making errors in different ways, diverse models can correct for each other when their probability estimates are combined, reducing the overall log-loss and producing better-calibrated predictions. This principle is especially pertinent in AutoML systems, where each phase in the

pipeline can shape both the individual model’s calibration and the ensemble’s collective calibration.

Recent theoretical work (e.g., [15, 19]) underscores how diversity among ensemble members can enhance generalization performance for a variety of loss functions, including log-loss. Because log-loss is proper, it punishes poorly calibrated predictions more severely, thus rewarding ensembles that offer accurate probability estimates. Concretely, these analyses show that an ensemble’s expected log-loss can be bounded above by the average log-loss of the individual models minus a diversity term. This underscores the practical payoff of improving calibration via diverse predictive behaviors.

Building on these insights, our goal is to investigate how log-loss can be leveraged to understand the influence of each phase—meta-learning, Bayesian optimization, and final ensemble construction—in shaping well-calibrated probability estimates. Specifically, we ask: (i) Why is it beneficial to separate the AutoML process into multiple phases if our primary aim is lowering log-loss? (ii) In what ways do each of these phases foster better-calibrated predictions? (iii) How do these stages together produce an ensemble that exhibits improved calibration? (iv) Do the methods at each stage explicitly target diverse, well-calibrated models, or is enhanced calibration a byproduct of their individual improvements?

To answer these questions, we next describe our methodology for measuring log-loss and capturing the diversity of probability estimates at different phases within the AUTO-SKLEARN AutoML pipeline.

4.1 Log Loss and Diversity Measures

Ensemble methods enhance performance by combining multiple models, and a crucial factor in their success is the diversity among individual predictors. In this work, we will leverage the tools introduced in [16, 19] to precisely measure the diversity of an ensemble. Given an ensemble $\Theta = \{h_1, \dots, h_K\}$, we define the predictive distribution of the ensemble as:

$$p_{\Theta}(y | x) = \frac{1}{K} \sum_{i=1}^K p_{h_i}(y | x), \quad (1)$$

where $p_{h_i}(y | x)$ represents the probability assigned to class y by an individual model h_i . The expected log loss for a single predictor h_i is given by:

$$L(h_i) = \mathbb{E}_{\nu}[-\log p_{h_i}(y | x)], \quad (2)$$

where ν is the data-generating distribution. Similarly, the log loss of the ensemble is:

$$L(\Theta) = \mathbb{E}_{\nu}[-\log p_{\Theta}(y | x)]. \quad (3)$$

To quantify diversity in ensembles using log loss, we employ a scaled variance measure defined by [16]. Instead of measuring diversity in terms of disagreement

on hard predictions, we analyze variations in predictive probabilities. More precisely, the diversity between two models h_i and h_j is then defined as:

$$d(h_i, h_j) = \mathbb{E}_\nu \left[\frac{p_{h_i}(y|x)^2 - p_{h_i}(y|x)p_{h_j}(y|x)}{2 \max_{k \in \{i,j\}} p_{h_k}(y|x)^2} \right]. \quad (4)$$

This measures how much the predictive probabilities differ between models, normalized by the maximum squared probability. When all models assign identical probabilities, the diversity is zero, while higher differences in predictive behavior increase the diversity score.

The total diversity of an ensemble, denoted $D(\Theta)$, is computed as the average diversity across all model pairs:

$$D(\Theta) = \frac{1}{K^2} \sum_{h_i, h_j \in \Theta} d(h_i, h_j). \quad (5)$$

Following [19, Theorem 1], we can establish an upper bound on the ensemble’s expected log loss:

$$L(\Theta) \leq \frac{1}{K} \sum_{i=1}^K L(h_i) - D(\Theta), \quad (6)$$

This inequality reveals two key insights: The first term represents the average log loss of individual models, showing their accuracy and calibration while the second term highlights the impact of diversity: greater diversity leads to better ensemble performance by improving uncertainty quantification.

This formulation provides a theoretical basis for optimizing AutoML ensembles for robust uncertainty estimation. The log loss and diversity measures discussed apply specifically to uniform probability averaging but they can also be extended to weighted probability averaging as shown in [16, 19].

5 Experimental framework and result analysis

Figure 1 shows a box-plot with a comparison of the distribution, across all datasets, of the error rate of base-learners and the error rate of the ensemble created by AUTO-SKLEARN (left image), and an analogous box-plot for log-loss (right image). The plot on the left shows that if we consider the error of the predictions of the raw material from which AUTO-SKLEARN departs (base-learners) and compare it with the model (ensemble) that results from the application of its different phases, the improvement is quite significant. Interestingly, although AUTO-SKLEARN has not been designed to optimize the calibration of the probabilistic estimates of its resulting ensemble, the figure on the right shows that the same mechanisms also serve to improve the quality of the probability predictions of the resulting ensemble, as the log-loss, which measures the quality of these probability predictions, is lower for the ensemble than for the individual models used to build it. Having said this, the aim of the experimentation

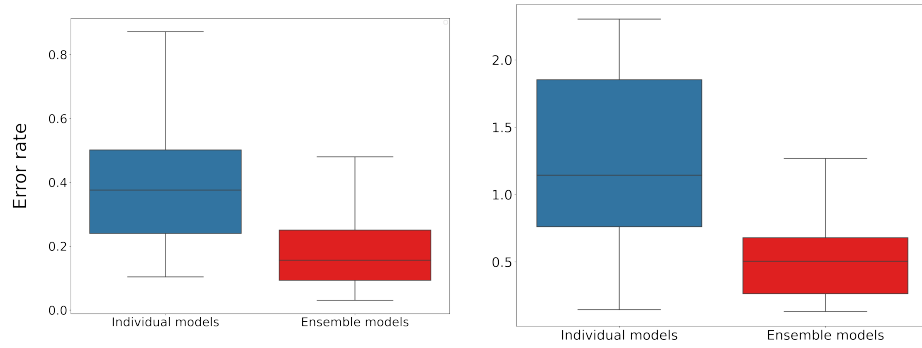


Fig. 1: On the left, box-plot with a comparison of the distribution, across all datasets, of the error rate of base-learners and the error rate of the ensemble created by AUTO-SKLEARN. On the right, analogous figure for log-loss.

carried out in this work is to analyze how the quality of the estimations of the models’ probabilities evolves through the different phases of AUTO-SKLEARN to achieve this improvement, and what role *diversity* plays in this improvement. To this end, in the next section we describe the experimental framework, while in the following section, the results of the experimentation are analyzed.

5.1 Experimental framework

We used 20 **datasets** from the AutoML benchmark [7] for our research, including four multi-class classification tasks and 16 binary classification tasks. This selection covers a wide range of domains, including text classification, image recognition, healthcare, and finance, ensuring a comprehensive representation of real-world challenges. The datasets offer a strong testing ground for AutoML methods since they differ in size, complexity, number of features, and class distributions. Each dataset is preprocessed and standardized for consistency, with metadata such as the number of instances, features, missing values, and class imbalance levels included for evaluation. The datasets utilized are described in Table 1.

In our analyses, we examine how diversity evolves across the **AUTO-SKLEARN framework’s four primary stages**, as depicted in Figure 2: Base-Learners, Meta-Learning, Bayesian Optimization, and the Caruana Ensemble. To determine how each pipeline contributes to the overall performance of the ensemble, we measure the diversity and log loss of the pipelines produced at each stage. We aim to assess the framework’s effectiveness in enhancing probability estimates and diversity by comparing these measures across stages. Our empirical analysis aims to **1)** Ensure that results are comparable across all stages and **2)** Measure log loss and diversity of classifiers at each stage. Below, we describe in more detail how the analysis is performed at each stage:

- **Base-Learners:** In the offline phase, 121 pipelines are generated.

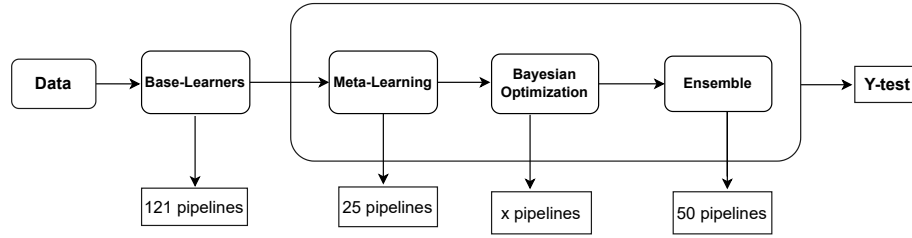


Fig. 2: Stages have been constructed in order to conduct experiments. The first stage depicts the 121 pipelines that exist in the offline stage of meta-learning. The second stage displays the 25 pipelines proposed to the Bayesian optimization by the meta-learning stage. The third and last stages show the pipelines generated using Bayesian optimization and the 50 pipelines in the ensemble, respectively.

- **Meta-Learning:** From the base-learners, 25 pipelines are selected by a meta-learning scheme to warm-up the Bayesian optimization phase.
- **Bayesian Optimization:** Pipelines are refined through Bayesian optimization, adapting their hyperparameter configurations to the dataset.
- **Caruana:** The optimized pipelines are combined into an ensemble using Caruana’s method, with up to 50 pipelines.

All pipeline sizes and method parameters were set to AUTO-SKLEARN’s default values as recommended by the authors in [4]. At each stage, we compute the log loss, $L(h_i)$, and pairwise diversity, $d(h_i, h_j)$, for all pipelines at that stage. Additionally, a majority-vote ensemble is created using the pipelines at each stage, and its log loss, $L(\Theta)$, is calculated to evaluate overall performance.

The experimental methodology also involves a structured **data partitioning approach**. Initially, the data is divided into training and testing sets. The training set is further subdivided into training and validation subsets. This partitioning serves different stages of the AUTO-SKLEARN framework:

- Three stages make use of the training data: Bayesian, meta-learning, and base-learners.
- The validation subset is specifically reserved for the Caruana ensemble stage.

An average of 112 pipelines are validated during the base-learners stage because of possible pipeline failures, although this may vary depending on the dataset. These validated pipelines generate predictions against the true labels. For both base-learners and meta-learning stages, we analyze the pipelines recommended by AUTO-SKLEARN, with meta-learning specifically focusing on 25 recommended pipelines. Recommended pipelines are extracted and analyzed at the Bayesian and Caruana stages, utilizing AUTO-SKLEARN’s resampling strategy and callback function to maintain consistent data partitioning and analysis throughout the process.

Table 2: Descriptive Statistics for individual model Log Loss (Eq (2)) across the different stages for 20 datasets

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.24	0.90	0.69	0.87
Std	0.65	0.53	0.45	0.52
Min	0.15	0.15	0.11	0.11
25%	0.76	0.47	0.44	0.42
50%	1.14	0.80	0.62	0.85
75%	1.85	1.20	0.76	1.19
Max	2.30	2.30	2.30	2.19

Table 3: Descriptive Statistics for pipeline Diversity (Eq (4)) across the different stages for 20 datasets

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.04	0.04	0.03	0.04
Std	0.02	0.02	0.02	0.02
Min	0.01	0.01	0.00	0.00
25%	0.02	0.02	0.01	0.02
50%	0.03	0.03	0.03	0.05
75%	0.06	0.06	0.04	0.07
Max	0.07	0.07	0.07	0.07

Table 4: Descriptive Statistics for the Ensemble Log Loss (Eq. (3)) across the different stages for 20 datasets

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.82	0.81	0.51	0.54
Std	0.54	0.52	0.46	0.34
Min	0.22	0.22	0.12	0.13
25%	0.51	0.51	0.24	0.26
50%	0.66	0.66	0.36	0.50
75%	0.94	0.95	0.59	0.68
Max	2.25	2.20	2.28	1.60

5.2 Result analysis

In this section, we analyze the results of the experimental framework described above. Concretely, we will first examine the evolution of the log-loss across all the AUTO-SKLEARN stages, and after that, we will do the same for the diversity.

Log-Loss Evolution Figure 3 illustrates the results of our investigation into log-loss across four phases of AUTO-SKLEARN methodology on three representative datasets: King Rook vs Pawn, Sylvine, and Amazon Employee. In each figure on the left, four box plots show the log-loss values for all individual pipelines generated during each of the four phases. For instance, in the Caruana phase, which includes 25 individual pipelines, the red box plot displays the distribution of $L(h_i)$ values for those pipelines.

In addition, for each of these four phases, we created an *ensemble* through the combination of the available pipelines at that stage. The overall log-loss of each of these ensemble, $L(\Theta)$, is shown by distinct horizontal lines in Figure 3. By comparing these lines against the box plots, one can observe how combining multiple pipelines can enhance probability calibration and reduce log-loss.

Figure 4(a) provides a more global perspective by aggregating the pipelines' log-loss across the full set of 20 datasets. In particular, the box plots in Fig-

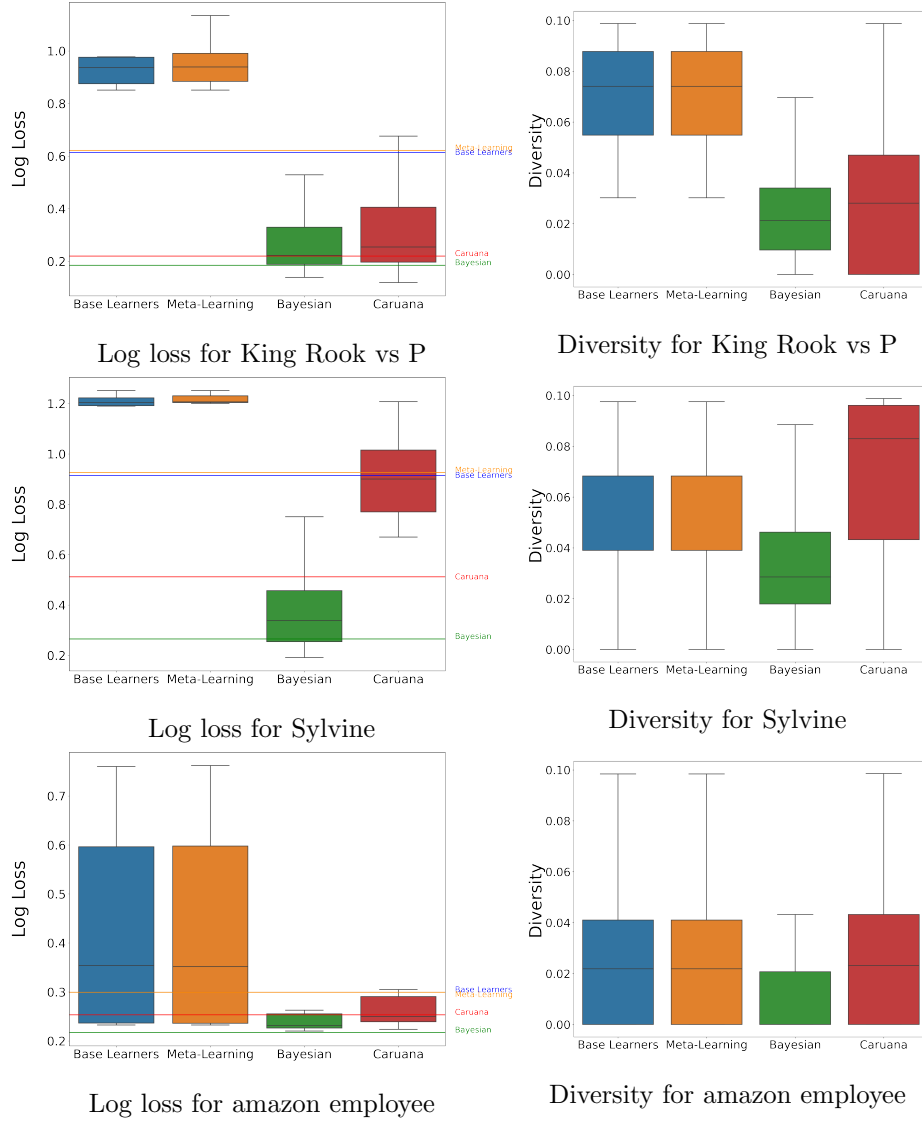


Fig. 3: Box-plot of the distribution of log loss and diversity across the four stages of AUTO-SKLEARN method for three datasets: King Rook vs Pawn, Sylvine, and Amazon Employee. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$, Equation (2) at each phase, while horizontal lines represent the overall ensemble log loss $L(\Theta)$, Equation (3), allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\Theta)$ is derived from Equation 5.

ure 4(a) reflect the average log-loss for each phase, computed as $\frac{1}{K} \sum_i L(h_i)$ thus highlighting the typical behavior of the pipelines at various stages. Meanwhile, Figure 4(c) summarizes the log-loss values of the ensembles built at the four phases, again over the entire set of 20 datasets. These two figures collectively give insight into how phases differ in terms of reliability and probability calibration.

From these results, the first two stages, *Base* and *Meta*, commonly exhibit higher log-loss. This aligns with the data in Table 2, where the *Meta-Learning* and *Base-Learners* phases show comparatively large median log-loss (e.g., 1.24 and 0.90). As a result, although these preliminary phases provide diverse models, their probability estimates are generally less calibrated.

In contrast, the *Bayesian Optimization* (BO) phase substantially reduces log-loss to an average of about 0.69 (Table 2), reflecting the positive impact of a more targeted and data-informed search for low-loss pipeline configurations. By exploiting training data to refine earlier pipeline choices, BO identifies more effective hyperparameter settings, leading to better-calibrated probability estimates.

Interestingly, individual pipelines introduced in the final, *Caruana* phase can sometimes yield higher log-loss than those from BO alone, having a mean of roughly 0.87 and a standard deviation of 0.52 (see Table 2). Yet, as indicated in Figure 4(c), the *ensemble* constructed at the Caruana stage performs comparably to the BO-based ensemble. This suggests that the Caruana method successfully draws upon a smaller set of 50 pipelines with sufficient diversity to create a well-calibrated final ensemble.

The Appendix A includes further plots that examine log-loss and diversity for 17 additional datasets, reinforcing the patterns noted here.

Diversity Evolution Figure 3 (plots on the right) illustrate the outcome of our diversity analyses for the same three datasets mentioned in the previous section. In each case, the box plots on the right side depict the distribution of pairwise diversity $d(h_i, h_j)$ among the individual pipelines produced in each of the four phases. We further compute the total ensemble diversity, $D(\Theta)$ by averaging over all possible pipeline pairs in that phase, thereby obtaining a single measure that captures how diverse the pipelines are within each ensemble.

Figure 4(b) provides a broader perspective by aggregating these diversity values for all 20 datasets. As seen there, the *Base* and *Meta* phases typically generate pipelines whose diversity is relatively high (mean around 0.04, as shown in Table 3). The *Bayesian Optimization* (BO) phase, in contrast, focuses on lowering individual log-loss but yields less diversity among its selected pipelines. The final phase, *Caruana*, then trends back toward higher diversity, as also evidenced by Figure 3 (right plots).

A natural question arises: *Why does the Caruana stage sometimes include pipelines with higher log-loss, and yet remain so competitive in terms of overall performance?* Figures 4(b)–(c) and Table 3 shed light on this apparent paradox. While the BO stage indeed finds pipelines with particularly low log-loss, it

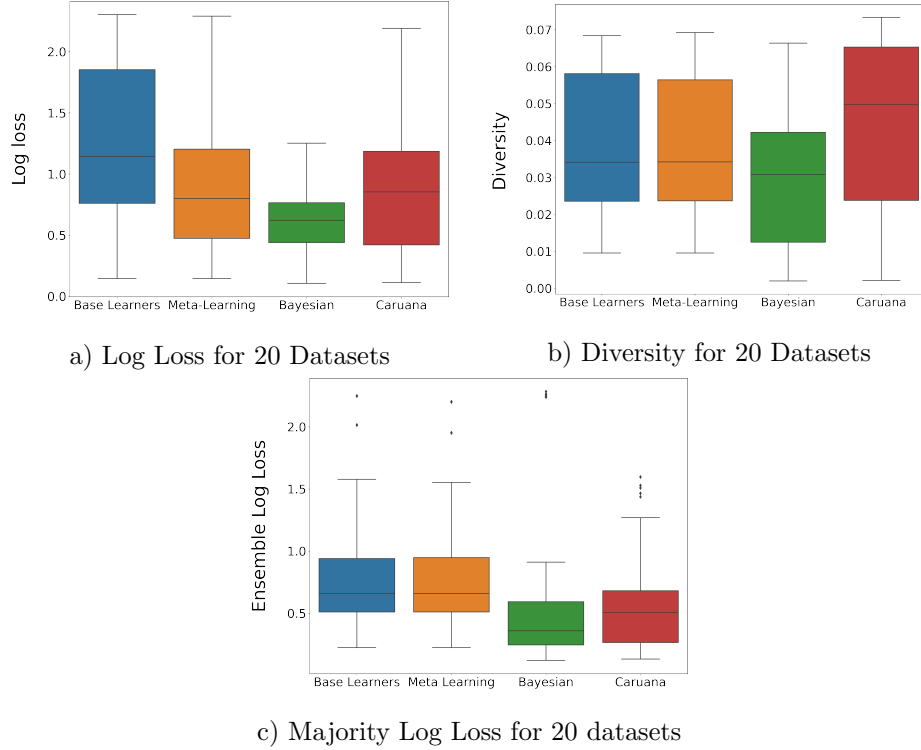


Fig. 4: Analysis of pipeline log loss and diversity across the four stages of the AutoML framework for 20 datasets: **(a)** Average log loss of individual pipelines at each stage, highlighting performance differences across stages. **(b)** Diversity of pipelines across the stages, similarly highlighting performance differences across stages. **(c)** Majority vote ensemble log loss for each stage, showcasing the combined impact of pipeline performance and diversity on ensemble effectiveness.

also tends to reduce diversity. The Caruana method, on the other hand, selects pipelines displaying greater diversity (with the highest mean diversity of 0.04 in Table 3), thereby improving ensemble-level calibration and generalization, even though some individual pipelines may exhibit higher log-loss values.

This outcome aligns closely with the theoretical arguments on ensemble generalization from Section 4.1, which emphasize that high ensemble diversity can offset moderate deficits in individual pipeline performance. In other words, the Caruana stage’s more diverse pipelines collectively achieve better probability calibration and lower overall log-loss than a less diverse ensembles. The consistently high diversity obtained by Caruana’s approach thus underpins its capacity to balance individual pipeline quality against the need for diverse prediction behaviors—ultimately leading to well-calibrated, accurate ensembles.

6 Conclusion

This paper has focused on the interplay between model diversity and well-calibrated probability predictions (measured in terms of log-loss performance) in ensemble-based Automated Machine Learning (AutoML) systems, using AUTO-SKLEARN as a case study. By synthesizing insights from both the theoretical foundations of uncertainty quantification and empirical evidence from the accomplished experimentation, we have demonstrated that diversity is as important as individual pipeline performance for achieving robust and well-calibrated ensembles.

Drawing on the analyses performed, our findings reinforce recent theoretical results showing that an ensemble’s overall log-loss and probability calibration depend on balancing the accuracy of individual models with the diversity of the ensemble. Through the phases of *Base-Learners*, *Meta-Learning*, *Bayesian Optimization*, and *Caruana* ensemble, AUTO-SKLEARN pursues a dynamic trade-off: some phases primarily improve individual log-loss, whereas others increase the variety of predictive behaviors.

Having said this, the specific answers to the four research questions raised are the following:

1. *Why is it beneficial to separate the AutoML process into multiple phases if our primary aim is lowering log-loss?* Our empirical analysis confirms that separate phases address different objectives. While some methods refine individual pipelines to minimize log-loss, other phases introduce or maintain diversity. Both objectives are necessary to build ensembles that are simultaneously accurate and well-calibrated.
2. *In what ways do each of these phases foster better-calibrated predictions?* Meta-learning provides a warm-start strategy but, in our experiments, did not substantially enhance individual log-loss or ensemble diversity. Bayesian Optimization excels at finding pipelines with notably lower log-loss; however, this refinement comes at the expense of decreased diversity. The Caruana phase compensates by including pipelines with higher log-loss yet significant

diversity, culminating in a final ensemble that is both accurate and well-calibrated.

3. *How do these stages together produce an ensemble that exhibits improved calibration?* Our observations show that different phases concentrate on each of these goals to varying degrees. Bayesian optimization is geared toward improving single-model performance, while Caruana’s phase explicitly leverages diversity. This synergy ensures that, as log-loss decreases, uncertainty quantification and probability calibration are preserved through diverse predictions.
4. *Do the methods at each stage explicitly target diverse, well-calibrated models, or is enhanced calibration a byproduct of their individual improvements?* Ultimately, AUTO-SKLEARN thrives on the interplay among these stages. Meta-learning sets an initial baseline, Bayesian Optimization systematically identifies better configurations, and the Caruana method provides a careful balance by selecting diverse yet individually suboptimal pipelines. The result is a final ensemble that combines accurate probability estimates with sufficient variability to improve generalization.

In summary, by uncovering the theoretical mechanisms through which ensemble-based AutoML enhances probability calibration, this work provides a foundation for the development of novel AutoML methodologies that prioritize not only predictive accuracy but also the reliability of uncertainty estimates. A key insight from our analysis is that diversity plays a crucial role in achieving well-calibrated probability estimates. Specifically, our findings show that any AutoML framework aiming to produce reliable probabilistic predictions should not only seek individual models with low log-loss but also ensure sufficient diversity among them. This balance between accuracy and diversity is essential for high-stakes applications, such as medical diagnosis, autonomous decision-making, and financial risk assessment, where miscalibrated probabilities can lead to suboptimal and potentially harmful outcomes. By leveraging these insights, future AutoML frameworks can be designed to produce models that offer both strong predictive performance and trustworthy probabilistic outputs, ultimately improving decision-making processes in critical real-world scenarios.

7 Acknowledgment

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References

1. Breiman, L.: Bagging predictors. *Machine learning* **24**, 123–140 (1996)
2. Brochu, E., Cora, V.M., De Freitas, N.: A tutorial on bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. *arXiv preprint arXiv:1012.2599* (2010)
3. Caruana, R., Niculescu-Mizil, A., Crew, G., Ksikes, A.: Ensemble selection from libraries of models. In: *Proceedings of the twenty-first international conference on Machine learning*. p. 18 (2004)
4. Feurer, M., Klein, A., Eggenberger, K., Springenberg, J.T., Blum, M., Hutter, F.: Efficient and robust automated machine learning. In: *Advances in neural information processing systems*. vol. 28, pp. 2962–2970 (2015)
5. Freund, Y., Schapire, R.E.: A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences* **55**(1), 119–139 (1997)
6. Gelbart, M.A., Snoek, J., Adams, R.P.: Bayesian optimization with unknown constraints. *arXiv preprint arXiv:1403.5607* (2014)
7. Gijbbers, P., LeDell, E., Thomas, J., Poirier, S., Bischl, B., Vanschoren, J.: An open source automl benchmark. *arXiv preprint arXiv:1907.00909* (2019)
8. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: *International conference on machine learning*. pp. 1321–1330 (2017)
9. He, X., Zhao, K., Chu, X.: Automl: A survey of the state-of-the-art. *Knowledge-based systems* **212**, 106622 (2021)
10. Kanter, J.M., Veeramachaneni, K.: Deep feature synthesis: Towards automating data science endeavors. In: *2015 IEEE international conference on data science and advanced analytics (DSAA)*. pp. 1–10. IEEE (2015)
11. Koza, J.R.: Genetic programming as a means for programming computers by natural selection. *Statistics and computing* **4**, 87–112 (1994)
12. Lakshminarayanan, B., Pritzel, A., Blundell, C.: Simple and scalable predictive uncertainty estimation using deep ensembles. In: *Advances in neural information processing systems*. pp. 6402–6413 (2017)
13. LeDell, E., Poirier, S.: H2o automl: Scalable automatic machine learning. In: *Proceedings of the AutoML Workshop at ICML*. vol. 2020 (2020)
14. Liu, H., Simonyan, K., Yang, Y.: Darts: Differentiable architecture search. *arXiv preprint arXiv:1806.09055* (2018)
15. Masegosa, A., Lorenzen, S., Igel, C., Seldin, Y.: Second order pac-bayesian bounds for the weighted majority vote. *Advances in Neural Information Processing Systems* **33**, 5263–5273 (2020)
16. Masegosa, A.R.: Learning under model misspecification: Applications to variational and ensemble methods. *Advances in Neural Information Processing Systems* **33**, 5479–5491 (2020)
17. Niculescu-Mizil, A., Caruana, R.: Predicting good probabilities with supervised learning. In: *International conference on machine learning*. pp. 625–632 (2005)
18. Olson, R.S., Moore, J.H.: Tpot: A tree-based pipeline optimization tool for automating machine learning. In: *Workshop on automatic machine learning*. pp. 66–74 (2016)
19. Ortega, L.A., Cabañas, R., Masegosa, A.: Diversity and generalization in neural network ensembles. In: *International Conference on Artificial Intelligence and Statistics*. pp. 11720–11743. PMLR (2022)

20. Pernot, P.: Calibration in machine learning uncertainty quantification: beyond consistency to target adaptivity. *APL Machine Learning* **1**(4) (2023)
21. Real, E., Aggarwal, A., Huang, Y., Le, Q.V.: Regularized evolution for image classifier architecture search. In: *Proceedings of the aaai conference on artificial intelligence*. vol. 33, pp. 4780–4789 (2019)
22. Snoek, J., Larochelle, H., Adams, R.P.: Practical bayesian optimization of machine learning algorithms. *Advances in neural information processing systems* **25** (2012)
23. Thornton, C., Hutter, F., Hoos, H.H., Leyton-Brown, K.: Auto-weka: Combined selection and hyperparameter optimization of classification algorithms. In: *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*. pp. 847–855 (2013)
24. Vanschoren, J.: Meta-learning: A survey. *arXiv preprint arXiv:1810.03548* (2018)
25. Yao, Q., Wang, M., Chen, Y., Dai, W., Li, Y., Tu, W., Yang, Q., Yu, Y.: Taking human out of learning applications: A survey on automated machine learning. *arXiv preprint arXiv:1810.13306* (2018)
26. Zhang, Y., Zame, W., van der Schaar, M.: Autocp: Automated pipelines for accurate prediction intervals. *arXiv preprint arXiv:2006.14099* (2020)
27. Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., He, Q.: A comprehensive survey on transfer learning. *Proceedings of the IEEE* **109**(1), 43–76 (2020)

Appendix A: Additional Visualizations

This appendix includes figures that offer deeper insights into the log loss and diversity distributions across the Base Learners, Meta-learning, Bayesian, and Caruana stages for each of the 20 individual datasets. These visualizations complement the analysis in Section 5 and 4, highlighting detailed comparisons and trends discussed in the main text.

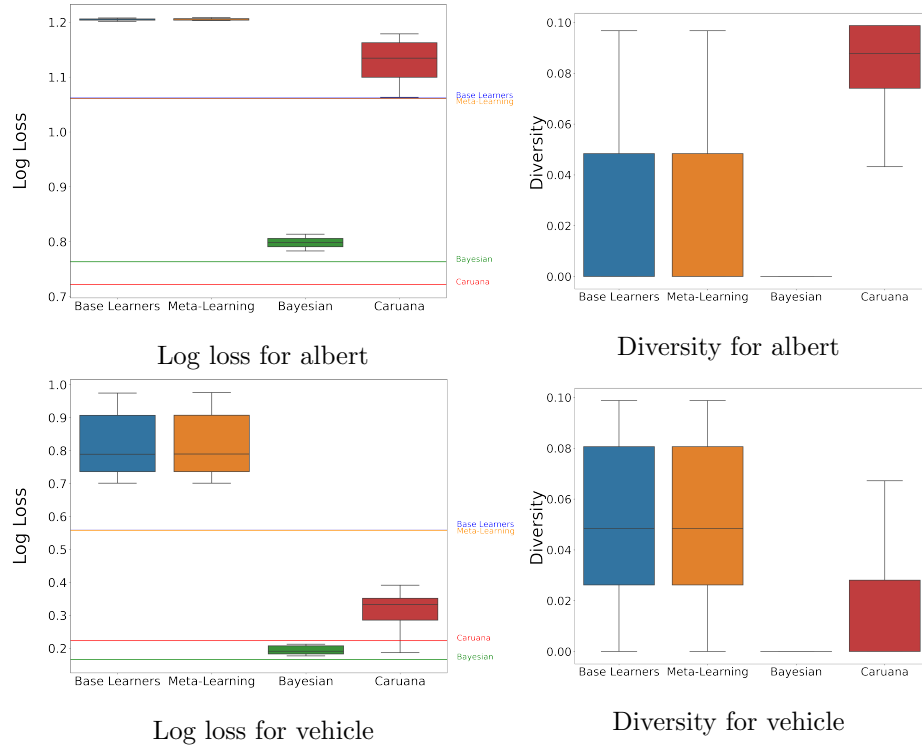


Fig. 5: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for albert, and vehicle. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\Theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\Theta)$ is derived from Equation 5.

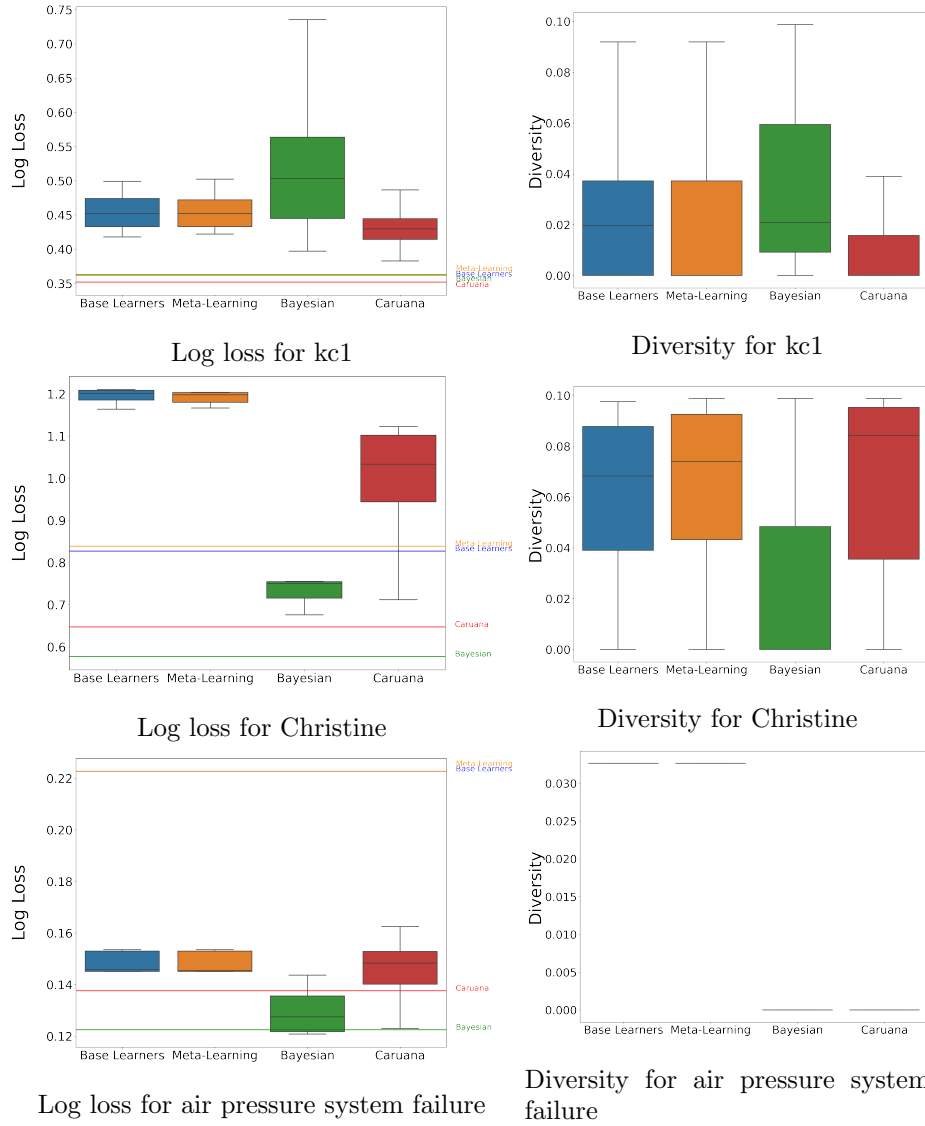


Fig. 6: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for kc1, Christine, and air pressure system failure. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\Theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\Theta)$ is derived from Equation 5.

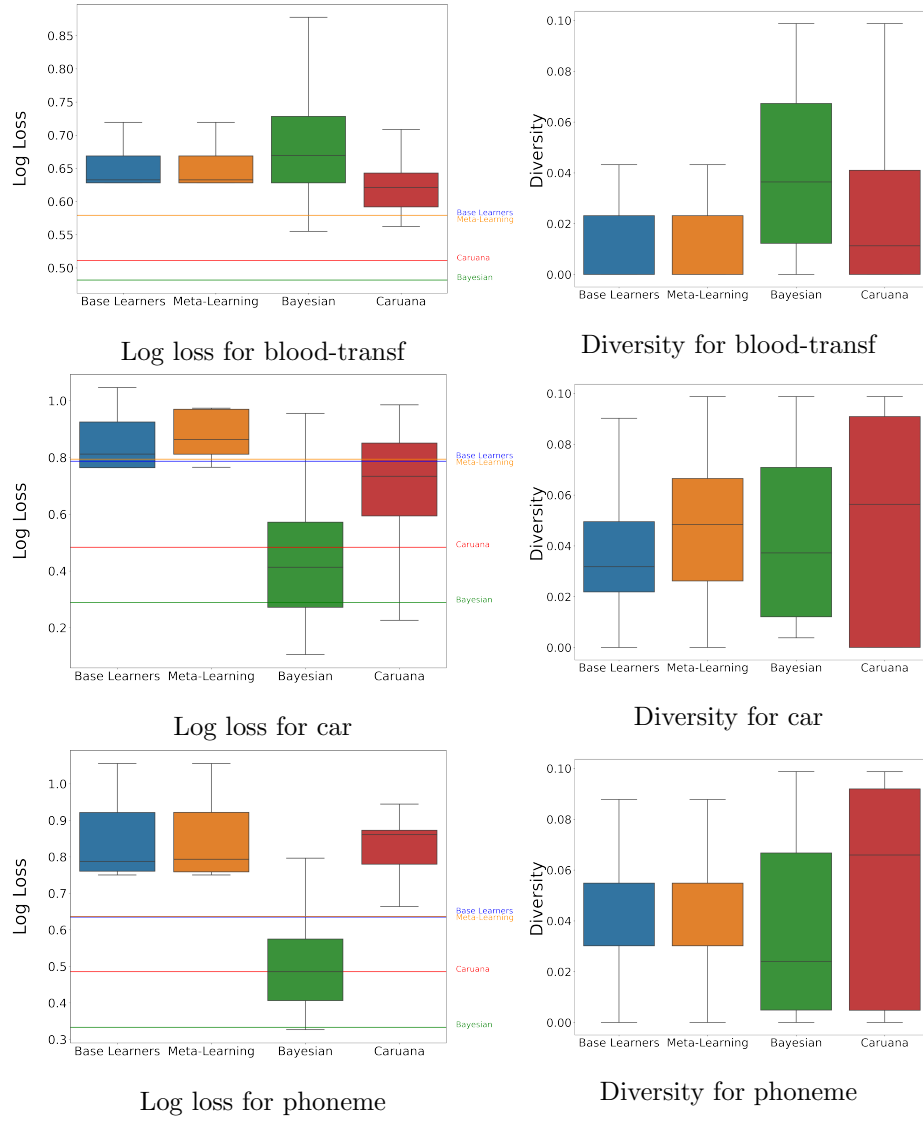


Fig. 7: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for blood-transf, car, and phoneme. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\Theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\Theta)$ is derived from Equation 5.

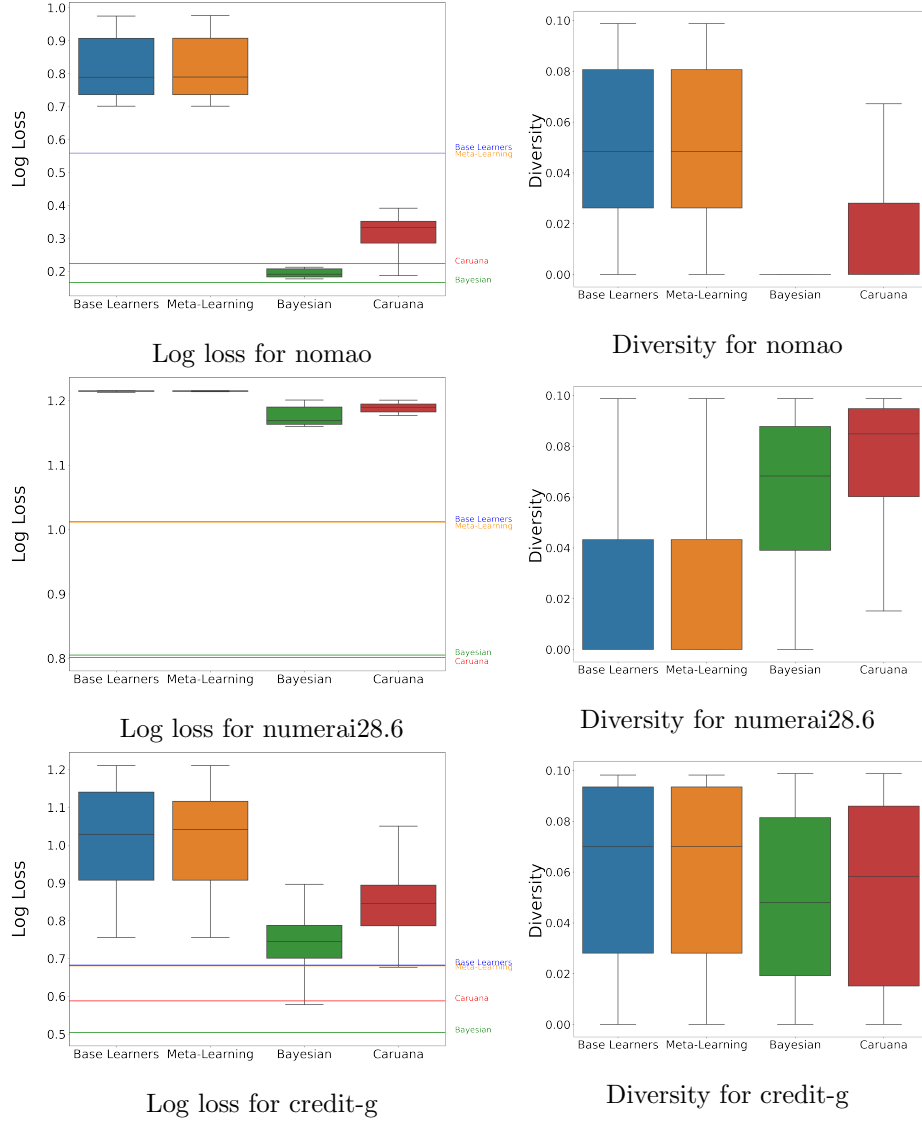


Fig. 8: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for nomao, numerai28.6, and credit-g. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\theta)$ is derived from Equation 5.

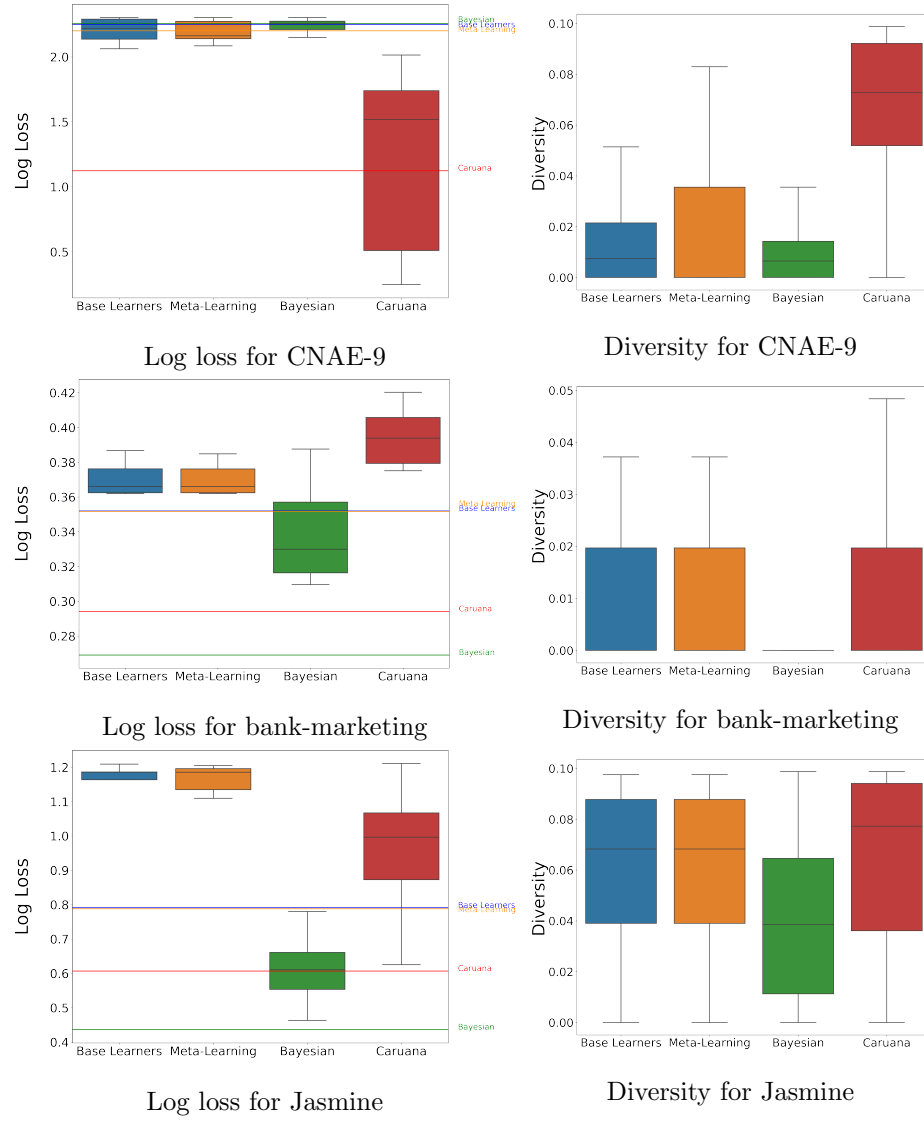


Fig. 9: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for CNAE-9, bank-marketing, and Jasmine. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\Theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\Theta)$ is derived from Equation 5.

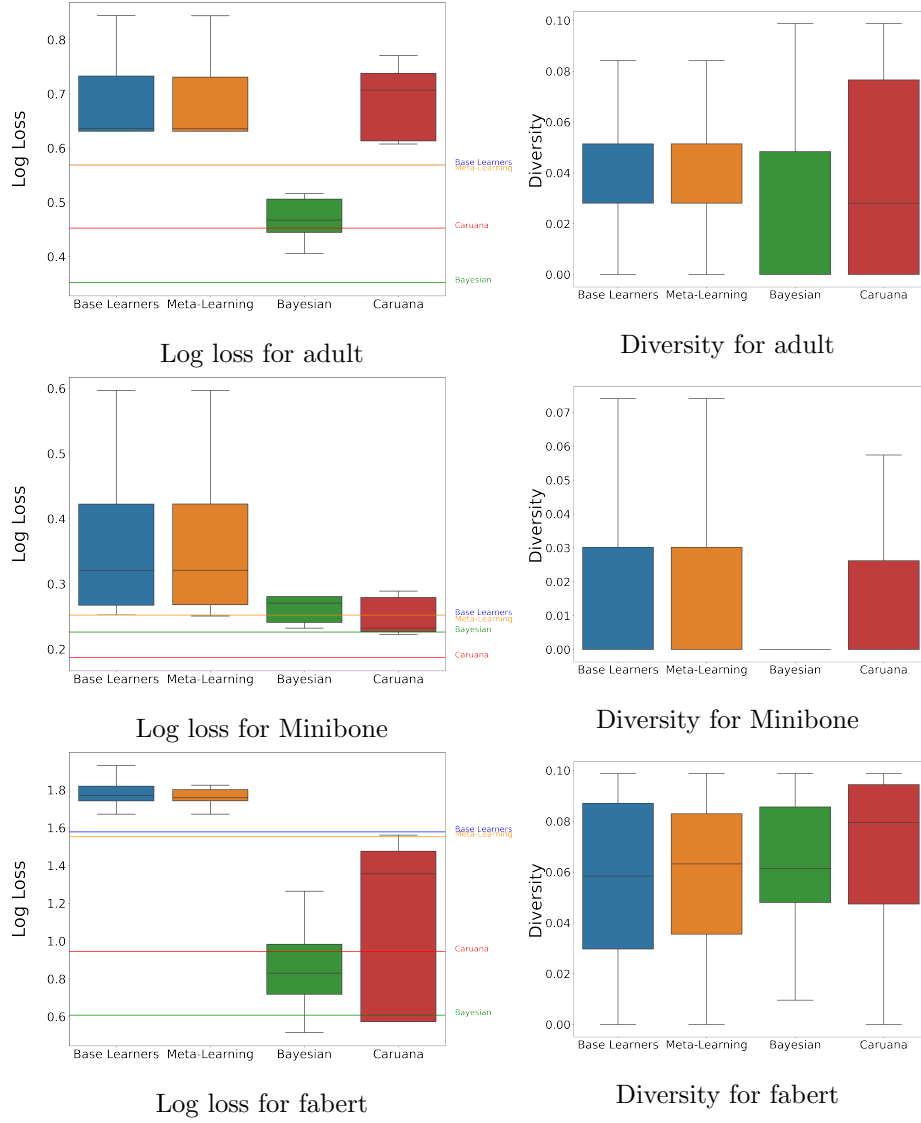


Fig. 10: The analysis examines log loss and diversity across the four stages of AUTO-SKLEARN method for adult, Minibone, and fabert. The box plots on the left illustrate the distribution of individual pipeline log loss $L(h_i)$: equation 2 at each phase, while horizontal lines represent the overall ensemble log loss $L(\theta)$: equation 3, allowing a comparison between individual pipeline performance and ensemble performance. The box plots on the right depict pipeline diversity $d(h_i, h_j)$, computed using Equation 4, while the total ensemble diversity $D(\theta)$ is derived from Equation 5.

Table 5: Descriptive Statistics for pipeline log loss for Blood-transf

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.69	0.69	0.69	0.63
Std	0.13	0.13	0.09	0.09
Min	0.63	0.63	0.56	0.56
25%	0.63	0.63	0.63	0.59
50%	0.63	0.63	0.67	0.62
75%	0.67	0.67	0.73	0.64
Max	1.12	1.12	0.99	1.20

Table 6: Descriptive Statistics for pipeline diversity for Blood-transf

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.02	0.02	0.04	0.03
Std	0.02	0.02	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.01	0.00
50%	0.02	0.02	0.04	0.01
75%	0.02	0.02	0.07	0.04
Max	0.10	0.10	0.10	0.10

Table 7: Descriptive Statistics for pipeline log loss for Sylvine

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.21	1.22	0.41	0.85
Std	0.02	0.02	0.22	0.26
Min	1.19	1.20	0.19	0.22
25%	1.19	1.20	0.25	0.77
50%	1.20	1.21	0.34	0.90
75%	1.22	1.23	0.46	1.02
Max	1.25	1.25	1.19	1.21

Table 8: Descriptive Statistics for pipeline diversity for Sylvine

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.05	0.05	0.03	0.07
Std	0.03	0.03	0.02	0.03
Min	0.00	0.00	0.00	0.00
25%	0.04	0.04	0.02	0.04
50%	0.04	0.04	0.03	0.08
75%	0.07	0.07	0.05	0.10
Max	0.10	0.10	0.10	0.10

Appendix B: Additional Analysis

Additional statistics for the 20 individual datasets focusing on log loss and diversity can be found below: These Tables (5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43) shows the statistics for the log loss and (6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44) represents statistics for diversity provides statistics the 20 individual datasets to complement Tables 2 and 3.

Table 9: Descriptive Statistics for pipeline log loss for Albert

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.21	1.21	0.80	1.11
Std	0.00	0.00	0.02	0.08
Min	1.20	1.20	0.78	0.90
25%	1.20	1.20	0.79	1.10
50%	1.20	1.20	0.80	1.13
75%	1.21	1.21	0.81	1.16
Max	1.21	1.21	0.81	1.18

Table 10: Descriptive Statistics for pipeline diversity for Albert

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.02	0.02	0.01	0.07
Std	0.03	0.03	0.02	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.00	0.07
50%	0.00	0.00	0.00	0.09
75%	0.05	0.05	0.00	0.10
Max	0.10	0.10	0.10	0.10

Table 11: Descriptive Statistics for pipeline log loss for jasmine

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.17	1.17	0.64	0.91
Std	0.04	0.04	0.15	0.20
Min	1.11	1.11	0.46	0.49
25%	1.16	1.13	0.55	0.87
50%	1.19	1.19	0.61	1.00
75%	1.19	1.20	0.66	1.07
Max	1.21	1.20	1.53	1.21

Table 12: Descriptive Statistics for pipeline diversity for jasmine

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.07	0.07	0.04	0.06
Std	0.03	0.03	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.04	0.04	0.01	0.04
50%	0.07	0.07	0.04	0.08
75%	0.09	0.09	0.06	0.09
Max	0.10	0.10	0.10	0.10

Table 13: Descriptive Statistics for pipeline log loss for Christine

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.20	1.20	0.74	0.99
Std	0.04	0.04	0.05	0.14
Min	1.14	1.14	0.68	0.70
25%	1.19	1.18	0.72	0.94
50%	1.20	1.20	0.75	1.03
75%	1.21	1.20	0.75	1.10
Max	1.29	1.27	0.85	1.12

Table 14: Descriptive Statistics for pipeline diversity for Christine

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.07	0.06	0.03	0.07
Std	0.03	0.03	0.03	0.04
Min	0.00	0.00	0.00	0.00
25%	0.04	0.04	0.00	0.04
50%	0.07	0.07	0.00	0.08
75%	0.09	0.09	0.05	0.10
Max	0.10	0.10	0.10	0.10

Table 15: Descriptive Statistics for pipeline log loss for Air Pressure System Failure

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.33	0.33	0.13	0.15
Std	0.61	0.61	0.01	0.01
Min	0.15	0.15	0.12	0.12
25%	0.15	0.15	0.12	0.14
50%	0.15	0.15	0.13	0.15
75%	0.15	0.15	0.14	0.15
Max	2.16	2.16	0.14	0.16

Table 16: Descriptive Statistics for pipeline diversity for Air Pressure System Failure

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.03	0.03	0.00	0.00
Std	0.01	0.01	0.01	0.01
Min	0.00	0.00	0.00	0.00
25%	0.03	0.03	0.00	0.00
50%	0.03	0.03	0.00	0.00
75%	0.03	0.03	0.00	0.00
Max	0.10	0.10	0.10	0.10

Table 17: Descriptive Statistics for pipeline log loss for KC1

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.47	0.47	0.53	0.43
Std	0.10	0.10	0.13	0.02
Min	0.42	0.42	0.40	0.38
25%	0.43	0.43	0.45	0.41
50%	0.45	0.45	0.50	0.43
75%	0.47	0.47	0.56	0.44
Max	0.87	0.87	1.36	0.49

Table 18: Descriptive Statistics for pipeline diversity for KC1

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.02	0.02	0.03	0.01
Std	0.03	0.03	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.01	0.00
50%	0.02	0.00	0.02	0.00
75%	0.04	0.04	0.06	0.02
Max	0.10	0.10	0.10	0.10

Table 19: Descriptive Statistics for pipeline log loss for Miniboone

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.36	0.36	0.28	0.26
Std	0.12	0.12	0.05	0.05
Min	0.25	0.25	0.23	0.22
25%	0.27	0.27	0.24	0.23
50%	0.32	0.32	0.27	0.23
75%	0.42	0.42	0.28	0.28
Max	0.60	0.60	0.36	0.41

Table 20: Descriptive Statistics for pipeline diversity for Dataset Mini-boone

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.02	0.02	0.01	0.02
Std	0.03	0.03	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.00	0.00
50%	0.00	0.00	0.00	0.00
75%	0.03	0.03	0.00	0.03
Max	0.10	0.10	0.10	0.10

Table 21: Descriptive Statistics for pipeline log loss for Phoneme

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.84	0.84	0.50	0.81
Std	0.10	0.10	0.14	0.12
Min	0.75	0.75	0.33	0.36
25%	0.76	0.76	0.41	0.78
50%	0.79	0.79	0.49	0.86
75%	0.92	0.92	0.57	0.87
Max	1.06	1.06	1.33	1.02

Table 22: Descriptive Statistics for pipeline diversity for Dataset Phoneme

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.04	0.04	0.04	0.05
Std	0.03	0.03	0.03	0.04
Min	0.00	0.00	0.00	0.00
25%	0.03	0.03	0.00	0.00
50%	0.03	0.03	0.02	0.07
75%	0.05	0.05	0.07	0.09
Max	0.10	0.10	0.10	0.10

Table 23: Descriptive Statistics for pipeline log loss for Nomao

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.85	0.85	0.20	0.31
Std	0.20	0.20	0.02	0.07
Min	0.70	0.70	0.18	0.16
25%	0.74	0.74	0.18	0.29
50%	0.79	0.79	0.19	0.33
75%	0.91	0.91	0.21	0.35
Max	1.49	1.49	0.25	0.39

Table 24: Descriptive Statistics for pipeline diversity for Dataset Nomao

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.06	0.06	0.01	0.02
Std	0.03	0.03	0.02	0.03
Min	0.00	0.00	0.00	0.00
25%	0.03	0.03	0.00	0.00
50%	0.05	0.05	0.00	0.00
75%	0.08	0.08	0.00	0.03
Max	0.10	0.10	0.10	0.10

Table 25: Descriptive Statistics for pipeline log loss for Numerai28.6

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.21	1.21	1.18	1.18
Std	0.00	0.00	0.02	0.01
Min	1.21	1.21	1.16	1.15
25%	1.21	1.21	1.16	1.18
50%	1.21	1.21	1.17	1.19
75%	1.22	1.22	1.19	1.19
Max	1.22	1.22	1.20	1.20

Table 26: Descriptive Statistics for pipeline diversity for Numerai28.6

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.03	0.03	0.07	0.07
Std	0.03	0.03	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.04	0.06
50%	0.04	0.04	0.07	0.08
75%	0.04	0.04	0.09	0.09
Max	0.10	0.10	0.10	0.10

Table 27: Descriptive Statistics for pipeline log loss for Bank-Marketing

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.40	0.40	0.36	0.38
Std	0.10	0.10	0.08	0.04
Min	0.36	0.36	0.31	0.30
25%	0.36	0.36	0.32	0.38
50%	0.37	0.37	0.33	0.39
75%	0.38	0.38	0.36	0.41
Max	0.78	0.78	0.55	0.42

Table 28: Descriptive Statistics for pipeline diversity for Bank-Marketing

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.01	0.01	0.01	0.02
Std	0.01	0.01	0.03	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.00	0.00
50%	0.00	0.00	0.00	0.00
75%	0.02	0.02	0.00	0.02
Max	0.10	0.10	0.10	0.10

Table 29: Descriptive Statistics for pipeline log loss for Adult

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.75	0.75	0.50	0.64
Std	0.29	0.29	0.09	0.13
Min	0.63	0.63	0.41	0.39
25%	0.63	0.63	0.44	0.61
50%	0.64	0.64	0.47	0.71
75%	0.73	0.73	0.51	0.74
Max	1.71	1.71	0.67	0.77

Table 30: Descriptive Statistics for pipeline diversity for Adult

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.04	0.04	0.03	0.04
Std	0.02	0.02	0.03	0.04
Min	0.00	0.00	0.00	0.00
25%	0.03	0.03	0.00	0.00
50%	0.03	0.03	0.00	0.03
75%	0.05	0.05	0.05	0.08
Max	0.10	0.10	0.10	0.10

Table 31: Descriptive Statistics for pipeline log loss for Amazon Employee

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.42	0.42	0.27	0.35
Std	0.19	0.19	0.12	0.22
Min	0.23	0.23	0.22	0.22
25%	0.24	0.24	0.23	0.24
50%	0.35	0.35	0.23	0.25
75%	0.60	0.60	0.26	0.29
Max	0.76	0.76	0.71	0.95

Table 32: Descriptive Statistics for pipeline diversity for Amazon Employee

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.03	0.03	0.01	0.03
Std	0.03	0.03	0.02	0.02
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.00	0.00
50%	0.02	0.02	0.00	0.02
75%	0.04	0.04	0.02	0.04
Max	0.10	0.10	0.10	0.10

Table 33: Descriptive Statistics for pipeline log loss for King-Rook vs King-Pawn

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.97	0.98	0.29	0.37
Std	0.13	0.15	0.17	0.28
Min	0.85	0.85	0.14	0.12
25%	0.88	0.88	0.19	0.20
50%	0.94	0.94	0.22	0.25
75%	0.98	0.99	0.33	0.41
Max	1.32	1.40	0.81	1.16

Table 34: Descriptive Statistics for pipeline diversity for King-Rook vs King-Pawn

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.07	0.07	0.03	0.03
Std	0.03	0.03	0.02	0.03
Min	0.00	0.00	0.00	0.00
25%	0.05	0.05	0.01	0.00
50%	0.07	0.07	0.02	0.03
75%	0.09	0.09	0.03	0.05
Max	0.10	0.10	0.10	0.10

Table 35: Descriptive Statistics for pipeline log loss for Credit-G

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.00	1.00	0.76	0.85
Std	0.16	0.16	0.09	0.08
Min	0.76	0.76	0.58	0.68
25%	0.91	0.91	0.70	0.79
50%	1.03	1.04	0.74	0.85
75%	1.14	1.12	0.79	0.89
Max	1.21	1.21	1.12	1.09

Table 36: Descriptive Statistics for pipeline diversity for Credit-G

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.06	0.06	0.05	0.05
Std	0.03	0.03	0.03	0.04
Min	0.00	0.00	0.00	0.00
25%	0.03	0.03	0.02	0.02
50%	0.07	0.07	0.05	0.06
75%	0.09	0.09	0.08	0.09
Max	0.10	0.10	0.10	0.10

Table 37: Descriptive Statistics for pipeline log loss for Car

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.89	0.97	0.44	0.71
Std	0.21	0.26	0.27	0.20
Min	0.76	0.77	0.11	0.14
25%	0.76	0.81	0.27	0.59
50%	0.81	0.86	0.41	0.73
75%	0.92	0.97	0.57	0.85
Max	1.70	1.65	2.03	0.99

Table 38: Descriptive Statistics for pipeline diversity for Car

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.04	0.05	0.04	0.05
Std	0.02	0.03	0.03	0.04
Min	0.00	0.00	0.00	0.00
25%	0.02	0.03	0.01	0.00
50%	0.03	0.05	0.04	0.06
75%	0.05	0.07	0.07	0.09
Max	0.10	0.10	0.10	0.10

Table 39: Descriptive Statistics for pipeline log loss for Vehicle

Statistic	Base	Meta	Bayesian	Caruana
Mean	1.78	1.76	0.86	1.17
Std	0.05	0.05	0.20	0.38
Min	1.67	1.67	0.52	0.57
25%	1.74	1.74	0.72	0.57
50%	1.77	1.76	0.83	1.36
75%	1.82	1.80	0.98	1.48
Max	1.93	1.82	1.73	1.56

Table 40: Descriptive Statistics for pipeline diversity for Vehicle

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.06	0.06	0.06	0.07
Std	0.03	0.03	0.02	0.03
Min	0.00	0.00	0.01	0.00
25%	0.03	0.04	0.05	0.05
50%	0.06	0.06	0.06	0.08
75%	0.09	0.08	0.09	0.09
Max	0.10	0.10	0.10	0.10

Table 41: Descriptive Statistics for pipeline log loss for CNAE-9

Statistic	Base	Meta	Bayesian	Caruana
Mean	2.21	2.19	2.23	1.25
Std	0.08	0.08	0.06	0.59
Min	2.06	2.08	1.97	0.25
25%	2.14	2.14	2.21	0.51
50%	2.22	2.16	2.26	1.52
75%	2.29	2.27	2.27	1.74
Max	2.30	2.30	2.30	2.01

Table 42: Descriptive Statistics for pipeline diversity for CNAE-9

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.01	0.01	0.01	0.06
Std	0.02	0.03	0.02	0.03
Min	0.00	0.00	0.00	0.00
25%	0.00	0.00	0.00	0.05
50%	0.01	0.00	0.01	0.07
75%	0.02	0.04	0.01	0.09
Max	0.10	0.10	0.10	0.10

Table 43: Descriptive Statistics for pipeline log loss for Fabert

Statistic	Base	Meta	Bayesian	Caruana
Mean	2.02	2.02	1.08	1.53
Std	0.04	0.03	0.23	0.28
Min	1.95	1.98	0.78	0.77
25%	1.98	1.99	0.95	1.44
50%	2.03	2.01	1.04	1.63
75%	2.05	2.05	1.24	1.71
Max	2.15	2.05	1.38	1.85

Table 44: Descriptive Statistics for pipeline diversity for Fabert

Statistic	Base	Meta	Bayesian	Caruana
Mean	0.03	0.02	0.06	0.07
Std	0.03	0.03	0.04	0.03
Min	0.00	0.00	0.00	0.00
25%	0.01	0.00	0.00	0.05
50%	0.02	0.00	0.06	0.07
75%	0.04	0.04	0.09	0.09
Max	0.10	0.10	0.10	0.10